Suiyang Guo

+86 158-9334-6463 | 1327217178@qq.com | Northwestern Polytechnical Univ. | MSc, Control Eng. | 2024.09—2027.06

MSc student in Control Engineering at Northwestern Polytechnical University (NPU), specializing in robotics and AI (graduating Jun 2027). Full-stack robotics engineer with end-to-end capability from mechanical/PCB design and STM32 firmware to ROS/ROS2, deep learning and host software. Currently a VLA (Vision-Language-Action) & World-Model algorithm intern at 视启未来 (incubated by IDEA Research / International Digital Economy Academy), training and deploying manipulation policies for a dual-arm robot. Participated in a national key research project; holds 6 patents (4 invention, 2 utility-model) and 2 software copyrights.

## Education

Northwestern Polytechnical University (NPU) |MSc, Control Engineering | Robotics & AI | GPA: 89.56/100 2024.09—2027.06

Zhengzhou University |BEng, Automation (Elite Class) | Robotics & AI | GPA: 3.54/4.0 2020.09—2024.06

## Technical Skills

- **Programming**: C/C++ (proficient), Python (fluent), MATLAB (fluent); Linux driver/application development
- **Robotics**: ROS/ROS2 (proficient); navigation, SLAM, motion planning, inverse kinematics
- **Embedded**: STM32/51/Arduino (proficient); development on IMX6ULL, STM32F4
- **Deep Learning**: PyTorch —image processing, speech recognition, CNN design
- **Hardware**: PCB design (Altium Designer, JLCPCB), 3D-printing modeling/slicing
- **Tools**: Git, PyQt5 (host software), Autodesk Inventor (3D modeling)

## Experience

**视启未来** (incubated by IDEA Research) |VLA & World-Model Algorithm Intern 2026.04—Present

- Trained a dual-arm cloth-folding policy on the open-source $\pi$0.5 (openpi) VLA; a curated-data + two-stage-init recipe cut action error (offline MAE@50) by about 77% vs. the prior best, and shipped to real-robot deployment
- Led VLA inference real-time optimization (RTC real-time chunking + Triton kernel autotuning), cutting single-step inference latency from about 256 ms to 32 ms (roughly 8x) and end-effector jitter by 81%; drove the ROS1 to ROS2 migration for on-robot serving
- Built an AWBC / RECAP-advantage offline policy-improvement pipeline; ran cross-embodiment co-training and weight-space model-merging experiments
- Brought up Tsinghua's X-VLA architecture on our platform, using firmware-level IK to cut the control cycle from 1.4 s to 0.33 s; built an offline vision-ablation gate that root-caused and fixed a dataloader bug which had made the model vision-blind
- Explored a self-supervised latent-milestone world model (LMVLA), injecting predicted sub-goals into the $\pi$0.5 action expert; evaluated on LIBERO / RoboTwin

Zhongqin Herui (Shaanxi) Technology Co., Ltd. |Navigation & Mapping Algorithm Engineer (Part-time) 2025.11—2026.01

- Developed 3D mapping and navigation for an orchard autonomous inspection vehicle
- Owned equipment/sensor selection, calibration, and sensor-data processing
- Built 3D maps from raw LiDAR, then occupancy grids for vehicle navigation

Huawei Digital Energy Bootcamp |Trainee (Training) 2025.07

- Studied digital-energy cloud microservice (DDD) architecture and AI applications/engineering in energy services
- Practiced prompt engineering and DevOps deployment workflows

Dongfeng Motor Co., Ltd. |Intern 2023.08

- Studied gasoline/diesel engine structure and principles; toured the full vehicle manufacturing process

## Projects

Series-Parallel Hybrid Wheeled Humanoid Robot |National Key Research Project | Key Provincial Lab —team member 2024.09—Present

- **Overview**: designed and built a series-parallel hybrid wheeled humanoid system —AGV base, parallel mechanism, two 6-DoF arms (in-house 6-axis / JAKA) and a 2-DoF head —for agricultural picking, teaching and industrial use
- **Stack**: Ubuntu 20.04, ROS Noetic, STM32F4, point-cloud processing, path planning, ShenTong DB
- **Work**: wrote the robot master-control program (3D mapping, path planning, object recognition, dynamic obstacle avoidance); point-cloud publishing, collision detection, IK; designed an in-house multi-DoF arm control scheme, doubling (100%) the 6-DoF arm control frequency; STM32F4 lower-level control; built a ShenTong database for high-concurrency CRUD

AI-LLM Voice-Interaction Robot |Outstanding Undergraduate Thesis | Solo developer 2023.12—2024.09

- **Overview**: a smart-home voice-chat robot combining NLP, LLMs and AI lip-sync
- **Stack**: Raspberry Pi 4B, Python, PyQt5, PyTorch, ROS, FreeRTOS
- **Work**: built the LLM interaction software (recording, speech recognition, LLM dialogue); speech synthesis + digital-human lip-sync; smart-home control; earned 2 software copyrights

TCM Therapy Device Series (3 generations) |"Challenge Cup" Provincial 2nd Prize | Core R&D member 2021.03—2023.09

- **Products**: smart gourd-moxibustion (gen 1) → warm-needle therapy (gen 2) → home mobile moxibustion (gen 3)
- **Stack**: STM32, FreeRTOS, fuzzy PID, sensor fusion, Bluetooth
- **Work**: led full-stack R&D across three generations (mechanical → circuit → embedded → host software); STM32 firmware (FreeRTOS, fuzzy-PID, sensor integration); hardware circuits (Bluetooth, servo drive, smoke handling); PCB design; multi-sensor (temperature/image/pressure) integration; servo & stepper motor drivers

## Awards & Honors

- "Challenge Cup" Henan Collegiate Academic Sci-Tech Works Competition (16th) Provincial 2nd Prize (2023)
- 2nd Collegiate Electrical & Electronic Engineer Innovation Contest Central-China 1st Prize (2023)
- 16th "Siemens Cup" Intelligent Manufacturing Challenge Western Region 2nd Prize (2022)
- Lanqiao Cup —Embedded Design & Development Provincial 3rd Prize (2022)
- HIT Zhengzhou Research Institute Training Camp Outstanding Camper (2022)
- Zhengzhou Univ. 2nd-Class Scholarship & Merit Student 2022—2023, 2024—2025

## Patents & Certificates

- 4 invention patents, 2 utility-model patents, 2 software copyrights
- CET-6 (College English Test Band 6): 473